

INITIALIZATION AND CALIBRATION

OVERVIEW

The AD9361 powers up into a sleep state for minimal power consumption. Before the AD9361 is operational, its clocks must be enabled and initial calibrations completed. The purpose of this section is to describe in detail the operation of the different initialization calibrations.

Table 1 lists the initialization calibrations documented in this section.

Table 1. Initialization Calibrations Detailed in this Section

Calibration	Run Frequency	Calibration Done Bit
BBPLL VCO calibration	Once; any time BBPLL frequency changes	0x5E[7], 1 when locked
RF synthesizer charge pump calibration	Once	Rx: 0x244[7], 1 when done Tx: 0x284[7], 1 when done
RF synthesizer VCO calibration	Occurs automatically when integer frequency word written. In TDD, occurs when TXNRX changes logic level.	Rx: 0x247[1], 1 when locked Tx: 0x287[1], 1 when locked
Baseband Rx analog filter tune	Once; update when BW changes	0x016[7], self clears when done
Baseband Tx analog filter tune	Once; update when BW changes	0x016[6], self clears when done
Baseband Tx secondary filter tune	Once; manual equations, update when BW changes	
Rx TIA calibration	Once; manual equations, update when BW changes	
Rx ADC setup	Once; manual LUT or equations, update when ADC sampling rate changes	
Baseband DC offset	Once	0x016[0], self clears when done
RF DC offset	Any time LO frequency changes more than 100 MHz	0x016[1], self clears when done
Rx quadrature calibration	Any time LO frequency changes more than 100 MHz	Tracking runs continuously
Tx quadrature calibration	Any time LO frequency changes more than 100 MHz	0x016[4], self clears when done

INITIALIZATION CALIBRATIONS

Initialization calibrations are calibrations that must be run each time the [AD9361](#) device is powered up or hard reset using the RESETB pin. Several of the calibrations only need to run once during initialization and do not re-run during operation.

Others are dependent on the carrier frequency, temperature, or other parameters and need to run initially and when certain events occur (such as changing the carrier frequency more than 100 MHz). As long as power is applied to the [AD9361](#) device, the calibration results are stored, including while in the SLEEP state.

The six calibrations in the calibration control are part of a calibration sequence state machine. After a calibration completes, the bit that was set to initialize the calibration will self clear. If more than one calibration is enabled in a single register write, the calibrations will progress in a set order controlled by a state machine in the [AD9361](#). Table 2 shows the sequence of calibrations. When the calibration sequence state holds a value of 0x1, the calibrations are complete. Some calibrations depend on the results of previously run calibrations. The Rx baseband filter and Tx baseband filter calibrations are not part of the calibration sequence state machine, and should run only when all other calibrations are not running.

Table 2. Automatic Calibration Sequence and Calibration Status

Calibration Sequence State[3:0]	Active Calibration
0x0	Calibration WAIT state
0x1	Calibrations done
0x2	Baseband DC offset calibration
0x3	RF Rx DC offset calibration
0x4	Tx1 quadrature calibration
0x5	Tx2 quadrature calibration
0x6	Rx1 quadrature calibration
0x7	Rx2 quadrature calibration
0x8	Tx monitor calibration (DC offset)
0x9	Rx GM\LNA gain step calibration
0xA to 0xF	Flush states

BASEBAND Tx SECONDARY FILTER

The baseband Tx secondary filter is a tunable single pole filter after the baseband Tx analog filter. The Tx secondary filter corner is ideally set to 5 times the baseband bandwidth to help filter out of band Tx noise emissions. This filter is programmed when the `ad9361_set_tx_rf_bandwidth` function is called.

Rx TIA CALIBRATION EQUATIONS

The Rx transimpedance amplifier (TIA) is located between the mixer and Rx baseband analog filter. The TIA has two gain settings (0 dB gain and -6 dB gain), and applies a single pole filter with a corner at $2.5\times$ the baseband bandwidth. The corner frequency of the Rx TIA is programmed when the `ad9361_set_rx_rf_bandwidth` function is called. When the gain index in the [AD9361](#) Rx gain table changes the Rx TIA gain setting, the [AD9361](#) device will automatically scale the amplifier to maintain the same corner frequency.

Rx ADC SETUP

The receive ADC is a third-order continuous time delta-sigma modulator and is highly programmable. The values in many of the ADC registers change with sampling clock frequency while others do not change but the correct initial settings are critical for proper operation. The ADC is programmed with the `ad9361_set_rx_rf_bandwidth` function.

BASEBAND DC OFFSET CALIBRATION

It is recommended to run the baseband DC offset calibration once during device initialization in the ALERT state. Since the

baseband signal path does not change with different wireless standards or clock frequencies, it should not need to be run again. The baseband DC offset correction values are stored for all of the Rx analog baseband filter gain steps. The correction words are applied as the Rx gain changes based on the current Rx gain table index. This calibration is run during the `ad9361_bb_dc_offset_calib` function call.

The one-shot baseband DC offset calibration will complete in a finite time depending on the device settings. The calibration time is 404,000 ClkRF cycles.

BASEBAND DC OFFSET TRACKING

Baseband DC offset tracking should be used in conjunction with the RF DC offset tracking option. A high-pass filter loop is utilized to track DC changes caused by the RF DC offset and Rx quadrature correction block. The speed and accuracy of the BB DC tracking loop are configurable. Also, an option exists for the BB DC offset tracking to attack/settle quickly after a gain change and then after a certain time switch to slower DC offset tracking updates. When using the tracking mode, first run an initial one-shot baseband DC offset calibration to minimize any static DC offsets in the signal chain. Then enable tracking along with the desired tracking loop settings. Baseband DC offset tracking is enabled in the `ad9361_tracking_control` function call.

RF DC OFFSET CALIBRATION

The RF DC offset calibration should be run once during initialization, or potentially when moving to a new carrier frequency that is more than 100 MHz away from the previous carrier frequency. This calibration should be run in the ALERT state while the Rx synthesizer is enabled. The calibration is run during the `ad9361_rf_dc_offset_calib` function. The internal calibration results LUT stores separate results for the RF Rx A input. If using the RF Rx B or C inputs along with the RF Rx A input, you should run the calibration twice, once with each input band selected. After calibrating each band, switching between from the A input to the B or C input should not require another calibration. Since the B and C inputs use the same calibration results, switching from Input B to Input C may require running the RF DC offset calibration.

When using the full Rx gain table, the RF DC offset calibration only calibrates at gain indexes that are designated to calibrate in the Rx gain table. This is because several consecutive gain steps may leave the front end gain at the same setting, while only changing baseband gain settings. The RF DC offset correction is only designed to remove DC offset due to the RF parts of the signal chain. By only running the calibration at gain indexes

that actually change the front end gain, the calibration time is reduced. If the LUT does not hold a DC correction value for the current Rx gain index, it will use the DC offset correction for the next higher gain index that was calibrated. In the case of a split Rx gain table, the calibration runs at each LMT gain index.

The RF DC offset tracking is enabled by with the `ad9361_tracking_control` function. The tracking triggers an RF DC offset update based on three events: Rx gain change, no energy detected, or when the ENSM exits the Rx state. The DC Offset Update bit field setting allows enabling or disabling any combination of these events. Disabling RF DC offset tracking would use the initially calibrated RF DC offset and never update the correction words.

The calibration time and accuracy is calculated using Equation 9 and Equation 10. The calibration begins at minimum gain (index = 0) and moves toward maximum gain. The RF DC offset correction for each enabled Rx gain index is stored in a LUT and applied when the Rx gain index is used. If the full Rx gain table is used, the RF DC offset is calibrated only at gain indexes specified to calibrate in the Rx gain table. If the Rx gain table is split, the Number of Enabled Gain indexes in Equation 10 equals 41 gain steps.

$$\text{Length Of Each Average} = \{(\text{RF DC Offset Count} \times 256) + 255\} + 32 \quad (9)$$

$$\text{Total Calibration time}_{(\text{CLKRF Clock Cycles})} = 21 \times \text{Length Of Each Average} \times (\text{Number of Enabled Gain indexes} + 1) \quad (10)$$

Rx QUADRATURE TRACKING CALIBRATION

The Rx quadrature tracking uses the Rx data to continuously minimize the phase and gain error in the receive path. The tracking algorithm is configured and enabled in the `ad9361_tracking_control` function. As soon as the AD9361 ENSM enters the Rx or FDD state, the tracking will begin minimizing the quadrature error.

Tx QUADRATURE CALIBRATION

The Tx quadrature calibration uses a calibration signal internally to minimize the Tx DC offset, gain, and phase errors to improve the performance of the transmit chain. The `ad9361_tx_quad_calib` function configures and runs this calibration during initialization in the ALERT state. Completion of the calibration can be monitored by reading the Calibration Control register until the appropriate bit self clears. When changing the carrier frequency, a much faster refresh calibration can be initiated if desired to update the Tx offset, gain and phase error corrections. It is also recommended to refresh the calibration results if the device temperature changes dramatically. The AuxADC can be used to measure the device temperature using the internal temperature sensor, and to know when to refresh the Tx quadrature calibration.

The Tx quadrature calibration is a convergence algorithm, but has a maximum calibration time described in the following paragraphs. Equation 11 calculates the number of CLKRF clock

cycles used for maximum calibration time. CLKRF is the clock rate at the output of the Tx FIR filter (after Tx FIR interpolation).

$$TX_{QuadcalTime(CLKRFclockcycles)} = \#TxChannels \times 94464 \quad (11)$$

The Tx quadrature calibration stores a separate set of calibration results for the Tx_A and Tx_B output paths. If using both the Tx_A output and the Tx_B outputs, run the calibration twice, once with each output path selected.

To setup the Tx quadrature calibration first make sure that the NCO frequency is within the Rx baseband filter bandwidth. Depending on the digital filter configuration, it may be necessary to sweep the NCO phase offset to find the optimal setting. More information on the setup of this phase sweep can be found in the Tx Quad Cal FAQ on the AD9361 Engineer Zone website.

If using a custom Rx gain table, verify that the Tx quadrature calibration gain index in register Tx quad full/LMT gain points to an index with the TIA index = 1 and LPF index = 0. When using a split gain table, Register Tx Quad Full/LMT Gain should point to a gain index with TIA index = 1. For a split gain table, set the LPF index in Register Tx Quad LPF Gain to a value of 0x00.